LoL Draft Analysis

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https://github.com/jcj1337/lol_draft_project

Introduction

League of Legends is and has been the world's most popular eSport for quite some time now; it is a 5v5 multiplayer game with nearly 200 characters with several unique interactions between each. As an avid player myself (reaching top 50 on the ladder a couple years ago!) I can attest to a natural player curiosity to estimate expected win percentage while loading into game (e.g. I look at the players on my team and the characters they are playing and heuristically judge my chances of winning). Deep learning further extends this idea by quantitatively interpreting the complex, non-linear interactions between characters that are difficult to capture with linear or rule-based models. I think by representing characters using learned embeddings and modeling interactions between all selected champions, neural networks can learn patterns of synergy and on the other end counter-play between the two team character drafts.

Illustration

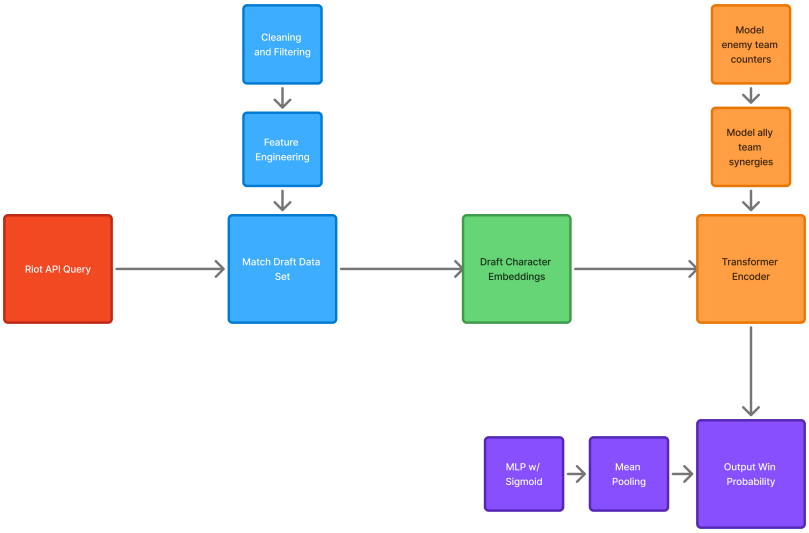

Figure 1: Model outline.

Background & Related Work

Prior work has explored predicting match outcomes in League of Legends with draft data, demonstrating that draft composition alone contains predictive signal [4]. More generally, the Riot API

has been explored to find key performance indicators at different stages of a game to predict win probability [6]. Beyond academic studies, data-driven analytics platforms such as LoLalytics provide aggregate champion statistics and matchup information derived from large-scale match data, while sites like OP.GG and u.gg serve as widely used sources of raw match and champion performance data for more casual players or analysts [1, 2, 3]. Furthermore, representation learning with embeddings has been shown to effectively model categorical entities in neural networks [5], and attention-based architectures such as transformers enable explicit modeling of interactions among multiple entities [7]. These past successes motivate the use of embedding-based, transformer models for a project like this one.

Data Processing

For this project I would be interested in making my own dataset. We will collect match data using the official Riot Games API to obtain character draft information and match outcomes. We should look to query "ranked solo/duo" games and extract, for each match, the ten selected champions (five per team) and the winning team label (0,1). To ensure reproducibility and reduce noise from balance changes, we will restrict our dataset to a fixed rank (masters+) and patch window, and we will store all raw API responses as JSON before converting them into a structured tabular format for training. So for a quick high-level view, we would have to query with an external API, do some data cleaning (e.g. remove duplicate match id's and verify 5 champions on each team) along with feature engineering with the raw data before training.

Architecture

0ur model at a high-level is taking in, for a given match, the 10 characters from draft and outputting a win percentage for both teams (should sum to 100, with "fair" drafts being closer to 50/50). Our input will be represented as champion IDs (0-171). Each champion ID is mapped to a learned embedding vector (e.g., 64–128 dimensions). We would then add a team embedding (blue vs. red) to each token so the model can distinguish allies from opponents. The resulting 10 vectors are processed by a transformer encoder (e.g., 2–4 layers) to model synergies (ally interactions) and counter-picks (opponent interactions) through self-attention. To obtain an actual output, we pool the transformer outputs (e.g. mean pooling) into a single vector. At this point we have done most of the work; to finish we can use some sort of means (like a MLP) to convert our singular vector into a scalar followed by a sigmoid to output a win percentage for blue side (of course 1-P(blue win) is P(red win)).

Baseline Model

We will use u.gg to help us create a hand-coded heuristic model. From u.gg, we can find data on champion-pair matchups for each of the 5 team roles. For each role, we obtain the win rate of the blue-side champion against the corresponding red-side champion. These five role-aligned matchup win rates can then be aggregated to produce a draft score for each team, and the predicted winner is the team with the higher total matchup score. This heuristic provides a simple baseline that incorporates a some level of interaction. I think this should be appropriate enough as a baseline given the complexity of the actual model is hard to capture entirely through a simpler means.

Ethical Considerations

Due to the use of an API, the dataset is subject to limitations based on how queries are structured. For instance, restricting queries to the North American region and applying rank thresholds introduces sampling bias, resulting in an exclusion of lower ranked players. Consequently, the findings may not be generalizable to the broader player population or to regions outside those queried.

Furthermore, the model does not account for underlying individual factors that can influence performance, such as mood, fatigue, or other external circumstances. Ignoring these variables may lead to inaccurate or oversimplified conclusions despite the size of the dataset.

References

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